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The Editors
Forensic Science International: Digital Investigation

Dear Editors,

We submit *Evidence-Monotone Audio Representation: Preventing Provenance Promotion Across Derived Audio Transformations* for consideration as a research article.

Current provenance standards can cryptographically bind assertions and transformation history to an asset. The missing layer addressed here is whether the evidence claim emitted by a derived representation is conservative with respect to every source interval that representation actually contains. A clip assembled from captured and generated material, then trimmed, resampled and re-encoded, can emerge described as captured-only, with a valid signature, an intact chain and a bit-identical waveform, because a derived asset inherits from its parent and a minimal implementation summarises that parent from the boundaries of the retained region. For an examiner reading such an asset, the difference between that claim and the truth is the whole point.

We contribute a formal operator contract for derived media, a temporal audio instantiation with declared kernel footprints and a footprint-monotonicity result that makes the rule implementable against real codecs, a signal-transparent reference implementation that requires no model training, a signed transport built entirely inside C2PA's own extension points, and a frozen public reproducibility artifact with one-command reproduction.

Four things are worth flagging to the editors directly, because we would rather state them than have a reviewer find them.

First, we do not claim the underlying algebra as new. The non-amplification property is an instance of a meet in a meet-semilattice, familiar from provenance semirings and from information-flow models, and we say so in the Introduction, the Formal model and the Discussion. The contribution is the executable temporal contract, the transport and the measurements.

Second, the comparison policy is a constructed reference policy motivated by the specification's own `parentOf` definition. It is not a description of any shipping product, no product is named or evaluated, and the fixture rates are reported as a mechanism stress rate rather than as a prevalence estimate.

Third, this is not a deepfake-detection paper. No model is trained, no acoustic feature is used, and an unsigned or provenance-stripped asset is classified as unverified rather than synthetic. A cryptographically valid signature proves that an assertion was signed, not that it is true, and we state that limitation in the abstract, the threat model, the discussion and the conclusion.

Fourth, the paper reports an independent reproduction that partly failed, and reports it as a result. A colleague at another institution, who is acknowledged but is not an author, ran the packaged release twice on unrelated hardware and a different FFmpeg major version. Every headline count returned unchanged and 10 of the 13 compared result files showed no deterministic difference at all. Two declared numbers did not hold on his build, and they are two different kinds: the MP3 kernel footprint is under-sized against the dependency his build actually has, and the silence-removal implementation margin is exceeded by the deviation between our temporal model and what his FFmpeg does. The reproduction tool returned a non-zero status and a verdict of `REPRODUCTION INCOMPLETE`, which the paper states rather than omits. We have not widened those declarations to make the comparison pass; Section 7.11 reports what he measured, and his unmodified outputs are released with the artifact.

The work fits the journal's stated primary pillar of digital evidence and multimedia with the core qualities of provenance, integrity and authenticity, and its "scientific practices" track: it is an attempt to make one specific property of derived digital evidence falsifiable and reproducible rather than assumed.

The manuscript is original, is not under consideration elsewhere, and all authors have approved the submission. A declaration of competing interests and a declaration of generative-AI use are included. All code, fixtures, corpora build scripts and machine-readable results are public under the MIT licence, with third-party data licences verified from primary sources.

Yours sincerely,

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